ECG Signal Denoising Using Wavelet Transform and Particle Swarm Optimization

INTRODUCTION

- Electrocardiogram (ECG) is an important signal used to monitor the heart and detect cardiovascular diseases.
- ECG signals are non-stationary
- ECG signal is often corrupted by noise like white Gaussian noise (WGN),power line interference (PLI)
- we are using Wavelet Transform (WT) for ECG signal denoising.
- To improve the denoising performance, we are using Particle Swarm Optimization (PSO)

PROBLEM STATMENT

- ECG signals are often corrupted by noise therefore it is necessary to denoise the signal while preserving the important information.

OBJECTIVE

- To Denoise the ECG signals using Wavelet Transform
- To use Particle Swarm Optimization (PSO) to find the optimal parameters of the Wavelet Transform.

WT IS DEFINED AS:

$$C(a,b) = \sum_{n \epsilon Z} x(n) g_{j,k}(n)$$

$$g_{j,k}(n) = 2^{j/2} g\left(2^{j} n - k\right)$$

$$a = 2^{-j}$$

$$b = k2^{-j}$$

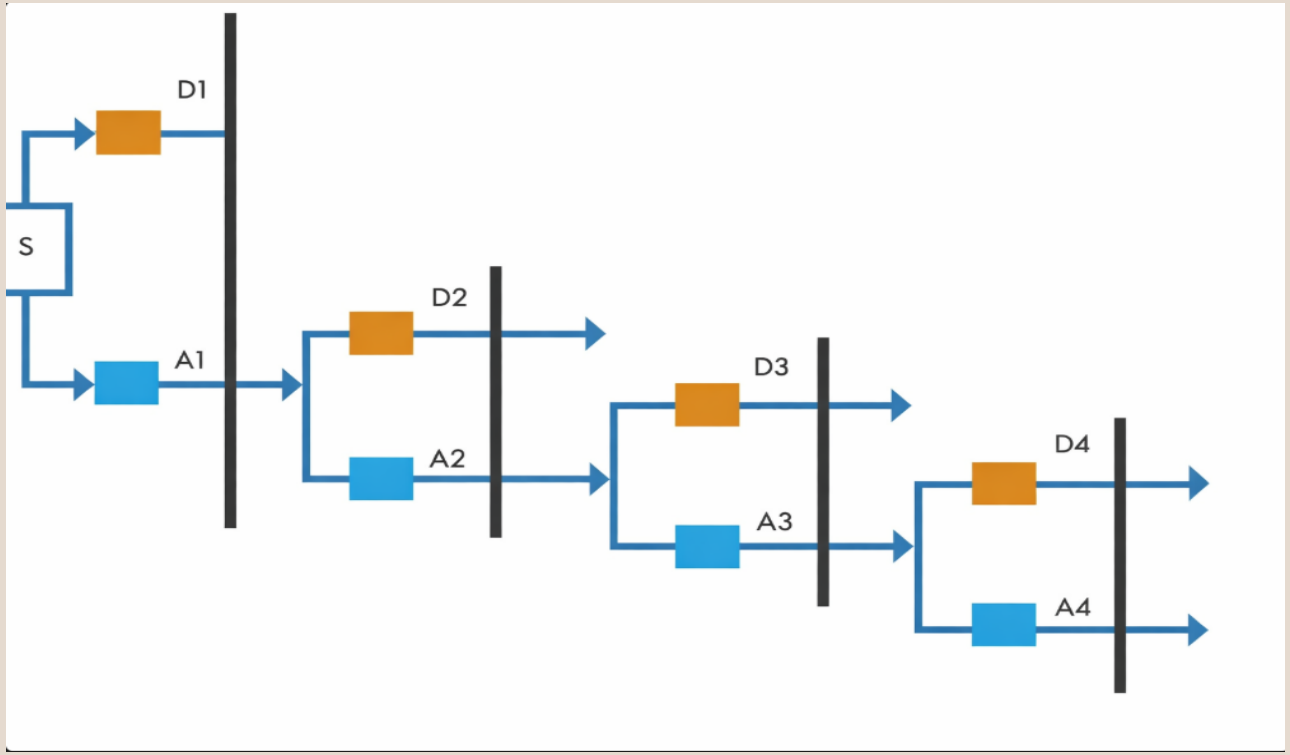
X(n)→original Signal

$g_{j,k}(n)$→Mother Wavelet

a→scaling

b→shifting

DWT



DWT

ECG Signal Decomposition:

- In wavelet decomposition, approximation coefficients contain the low-frequency information of the signal, whereas detail coefficients contain the high-frequency information.

Approximation coefficients (cA_i)

$$cA_i(t) = \sum_{k=-\infty}^{\infty} cA_{i-1}(K)\phi_i\,(t-K)$$

- cA_i(t): Approximation coefficients at decomposition level i
- cA_i −1(k): Approximation coefficients from the previous level
- $\phi_i(t)$:Wavelet function at level i
- K: summation index

Detail coefficients (cD_i)

$$cD_i(t) = \sum_{k=-\infty}^{\infty} cD_{i-1}(K)\psi_i\,(t-K)$$

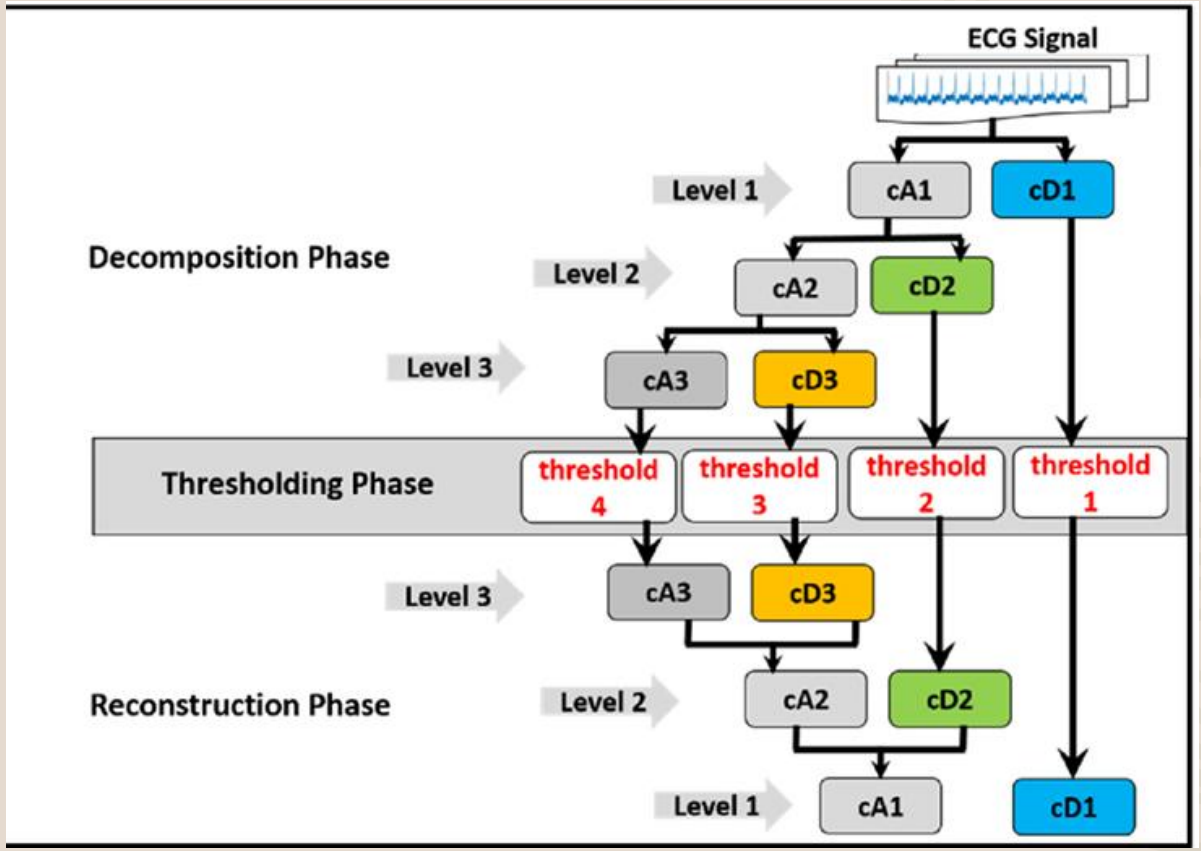
- $cD_i(t)$:Detail coefficients at decomposition level i
- $cD_i\ -1(k)$:Approximation coefficients from the previous level
- ψ_i *(t):* scaling function at level i
- K: summation index

Thresholding

$$\delta = \sigma(0.3936 + 0.1829 \log_2 N)\,, N > 32$$
$$0, \qquad N \leq 32$$

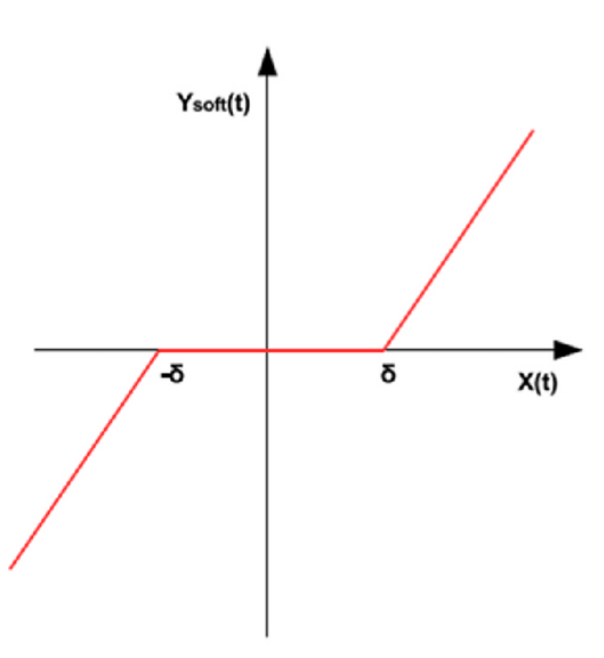
$$\sigma = \frac{median(|D_{ij}|)}{0.6745}$$

- N is the length of the signal vector.
- i → index of the coefficient
- j → decomposition level



Thresholding Function

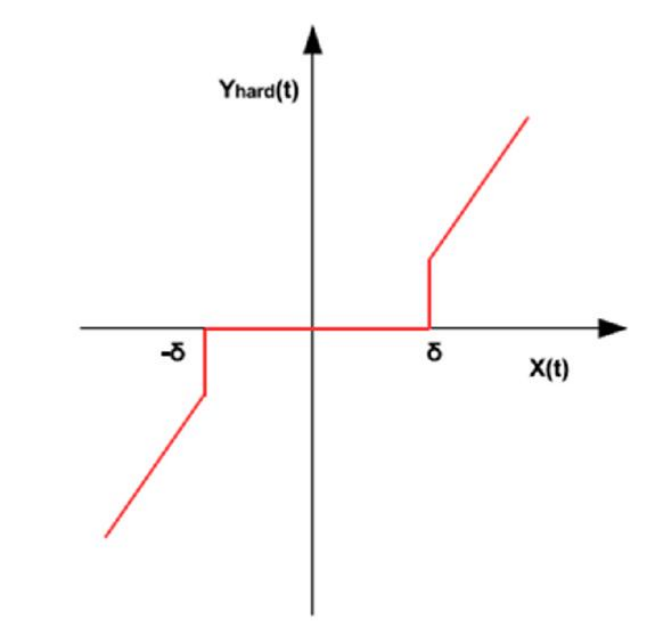
- Soft Thresholding



$$\widehat{x_{di}}(l) = \widehat{x_{di}}(l) \qquad |\widehat{x_{di}}(l)| > \delta_l$$

$$\widehat{x_{di}}(l) = 0 \qquad |\widehat{x_{di}}(l)| < \delta_l$$

- Hard Thresholding



$$\widehat{x_{di}}(l) = \widehat{x_{di}}(l) - |\delta_l| \qquad |\widehat{x_{di}}(l)| \geq \delta_l$$

$$\widehat{x_{di}}(l) = 0 \qquad |\widehat{x_{di}}(l)| < \delta_l$$

Reconstruction

$$\text{ECG_denoised} = \sum_{k=-\infty}^{\infty} cA_L(K)\phi_i(t-K) + \sum_{i=1}^{L} \sum_{k=-\infty}^{\infty} cD_{i+1}(K)\psi_i(t-K)$$

- $cA_L(k)$ → Approximation coefficients at final level L
- $\widehat{cD}_i(k)$ → Thresholded detail coefficients at level i
- ϕ(t) → Scaling (low-pass) function
- $\psi_i(t)$ →Wavelet (high-pass) function
- L → Total number of decomposition levels

Particle Swarm Optimisation

- PSO is an optimization algorithm used to find the best solution to a problem.
- It is inspired by the behavior of birds or fish moving in groups.
- When they search for food, they do not know the exact location.
- But one bird may find food first.
- The other birds observe its movement and move toward that area.
- In PSO, many possible solutions are created and called particles.
- A group of these particles is called a swarm.

•Particle → one possible solution

•Swarm → group of particles

$$x = [\phi, \beta, L, \lambda, \rho]$$

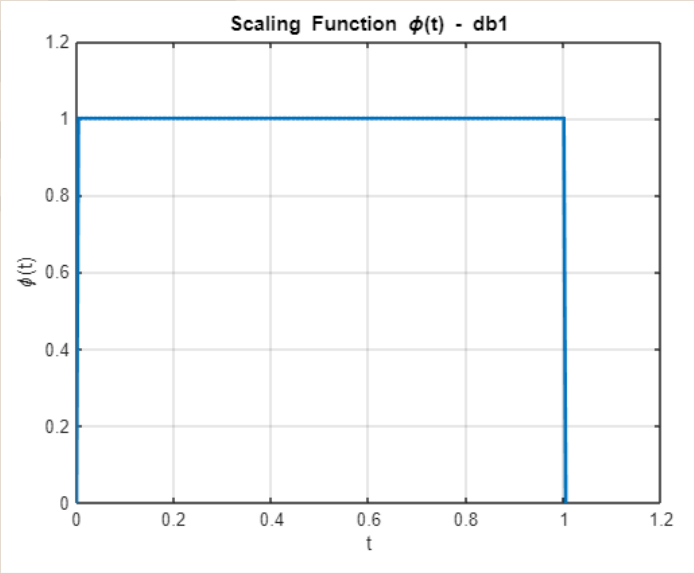
In Each particle we have 5 parameters such as

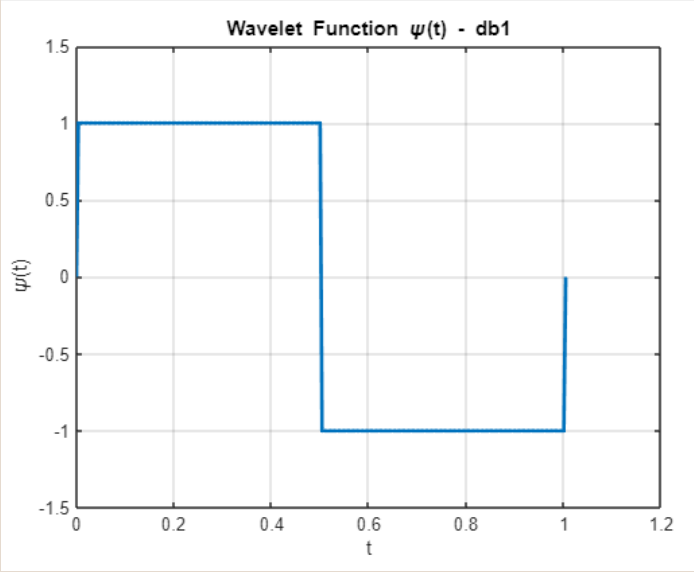
- Wavelet basis function (ϕ) - Coiflet , Symlet , Daubechies, Fejer-Korovkin and Biorthogonal
- Threshold function (β) – soft ,hard
- Level of decomposition (L) – 1 to 10
- Threshold selection rule (λ) - Sqtwolog, Minimax, Heursure, and Rigsure
- rescaling approach (ρ) - No scaling (one), one level (sln), Several levels (mln).

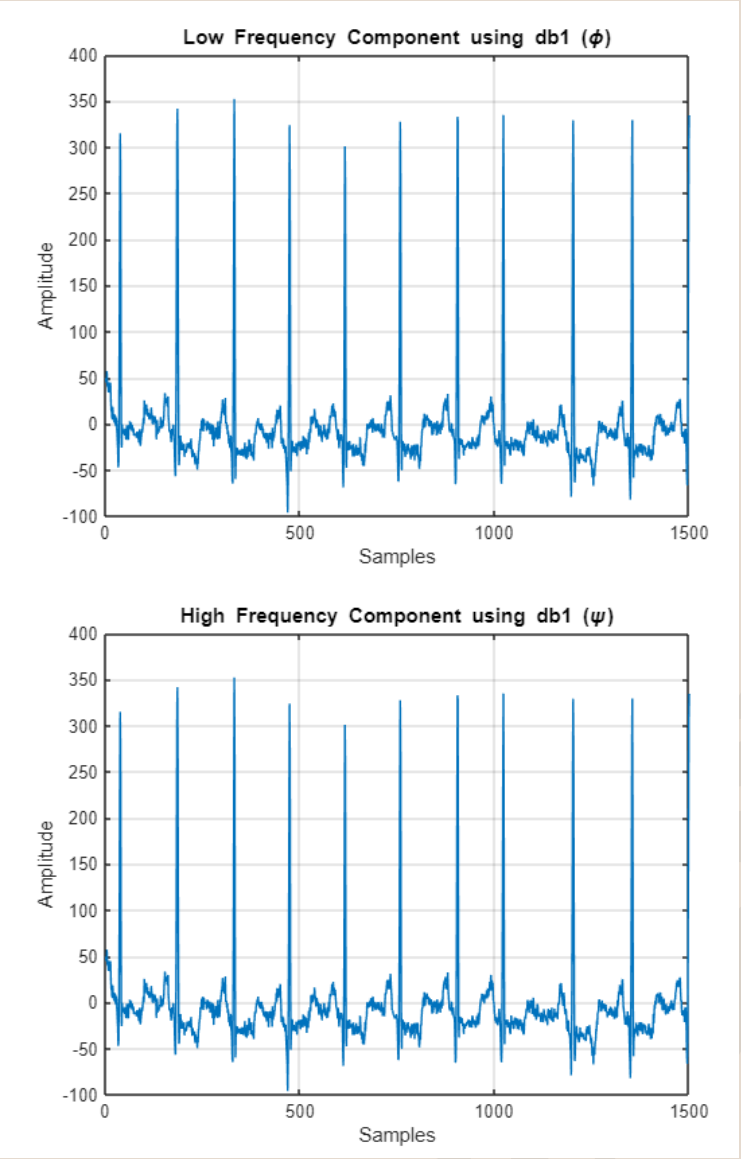
Threshold Selection Rules

Rule	**Simple Meaning**
Rigrsure	• Estimates the error between denoised and true signal • Chooses threshold that minimizes this error.
Sqtwolog	• The threshold formula: $$T = \sigma\sqrt{2\log M}$$ Where: σ= noise standard deviation M= number of coefficients.
Heursure	• Rigrsure + Sqtwolog. • If noise is small → use Rigrsure • If noise is large → use Sqtwolog.
Minimaxi	Minimize the maximum mean square error

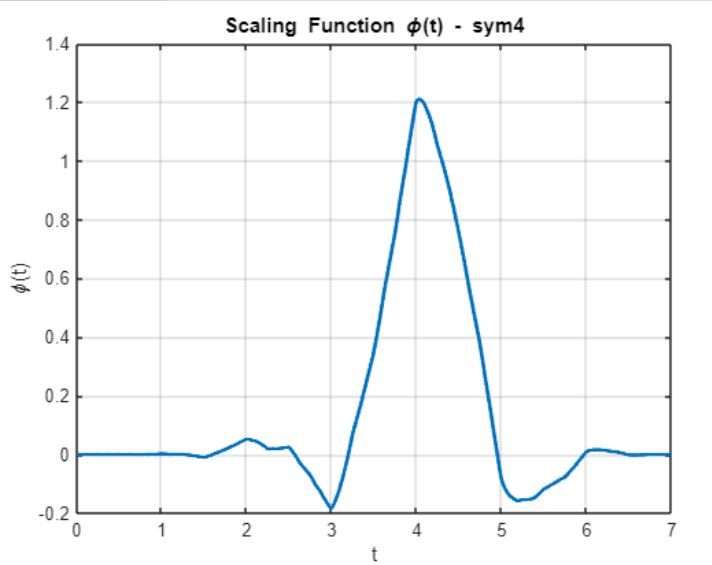
db1

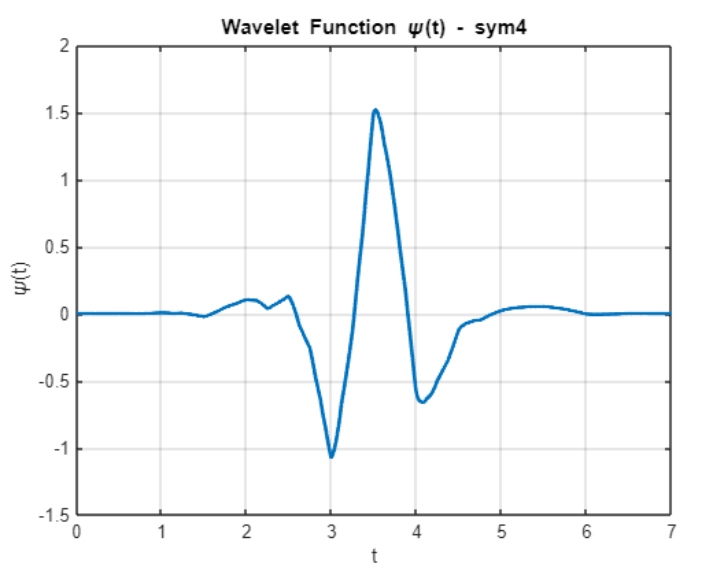

Scaling Function ϕ(t) - db1
ϕ(t)
t

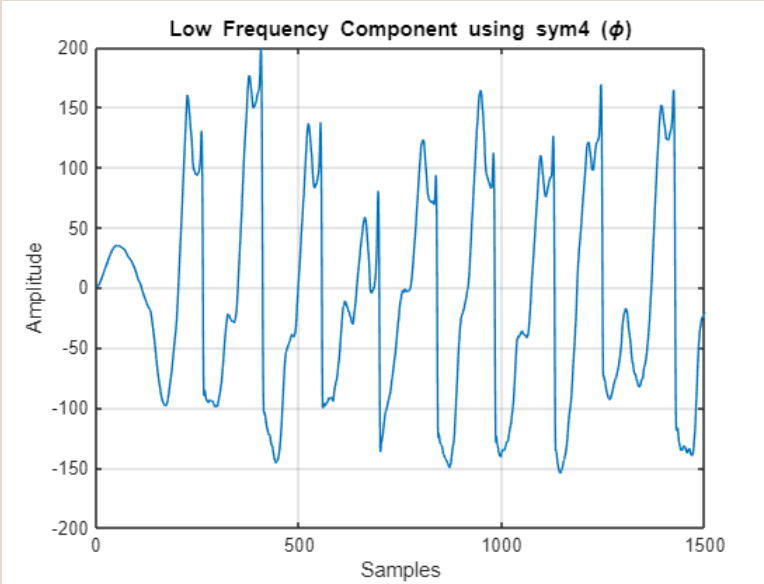

Wavelet Function ψ(t) - db1
ψ(t)
t

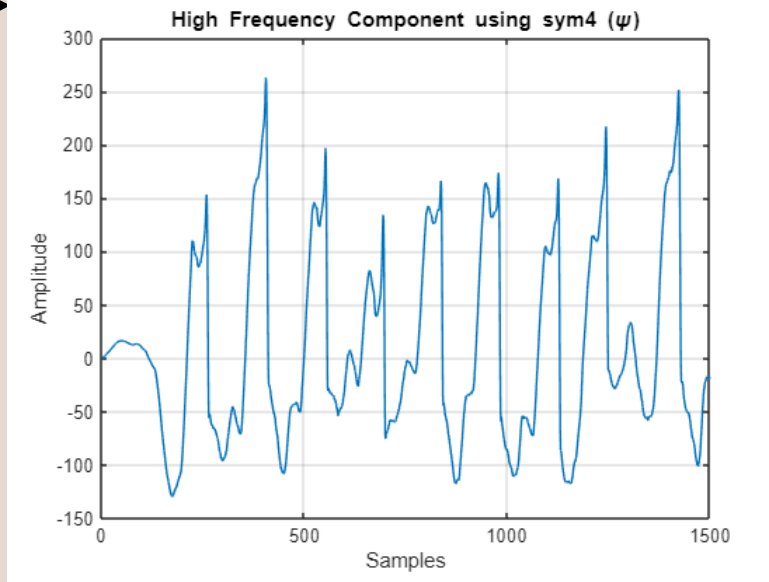

Low Frequency Component using db1 (ϕ)
Amplitude
Samples
High Frequency Component using db1 (ψ)
Amplitude
Samples

sym4


Scaling Function ϕ(t) - sym4
ϕ(t)
t


Wavelet Function ψ(t) - sym4
ψ(t)
t


Low Frequency Component using sym4 (ϕ)
Amplitude
Samples


High Frequency Component using sym4 (ψ)
Amplitude
Samples

Velocity

$$v_i(t+1) = wv_{i,j}(t) + c_1 r_{1,j}(t)\left(y_{i,j}(t) - x_{i,j}(t)\right) + c_2 r_{2,j}(t)\left(\widehat{y_j}(t) - x_{i,j}(t)\right)$$

$\left(y_{i,j}(t) - x_{i,j}(t)\right)$ → Local Best

$\left(\widehat{y_j}(t) - x_{i,j}(t)\right)$ → Global Best

- c1 and c2 are two acceleration constants
- w is inertia weight
- r1 and r2 generate a uniformly distributed random number between 0 and 1

Position

$$x_i(t+1) = x_i(t) + v_i(t+1)$$

Fitness:

$$SNR_{imp} = SNR_{output} - SNR_{input}$$

$$SNR_{input} = 10\log_{10}\sum_{n=1}^{N}[S(n)]^2 \ / \sum_{n=1}^{N}[S(n) - \tilde{S}(n)]^2$$

$$SNR_{output} = 10\log_{10}\sum_{n=1}^{N}[S(n)]^2 \ / \sum_{n=1}^{N}[S(n) - \hat{S}\ (n)]^2$$

$S(n)$→*Original Signal*

$\tilde{S}(n)$→ is the noisy ECG signal

$\hat{S}(n)$→ is the denoised ECG signal

Toy Example:

Index	Wavelet
1	coif1
2	coif3
3	sym3
4	sym5
5	db4
6	db6

Index	Threshold rule
1	Rigsure
2	Sqtwolog
3	Heursure
4	minimaxi

Index	Threshold
1	soft
2	hard

Index	Rescaling
1	one
2	sln
3	mln

Particle	ϕ	L	β	λ	ρ
P1	3	4	1	2	2
P2	5	6	1	1	2
P3	2	3	2	3	1

Particle	Pbest
P1	[3,4,1,2,2]
P2	[5,6,1,1,2]
P3	[2,3,2,3,1]

Particle	Velocity.L
P1	0
P2	0
P3	0

Particle	Fitness
P1	7
P2	9
P3	5

- gbest=P2
- Because of High Fitness
- w=0.7
- c1=2
- c2=2
- r1=0.4
- r2=0.6

Velocity =w*velocity+C1*r1*(pbest – particle)+C2*r2*(gbest - particle)

W*velocity=0.7*0=0
c1*r1*(pbest – particle)=2*0.4*(4-4)=0
c2*r2*(gbest - particle)=2*0.6*(6-4)=2.4
New Velocity is 2.4

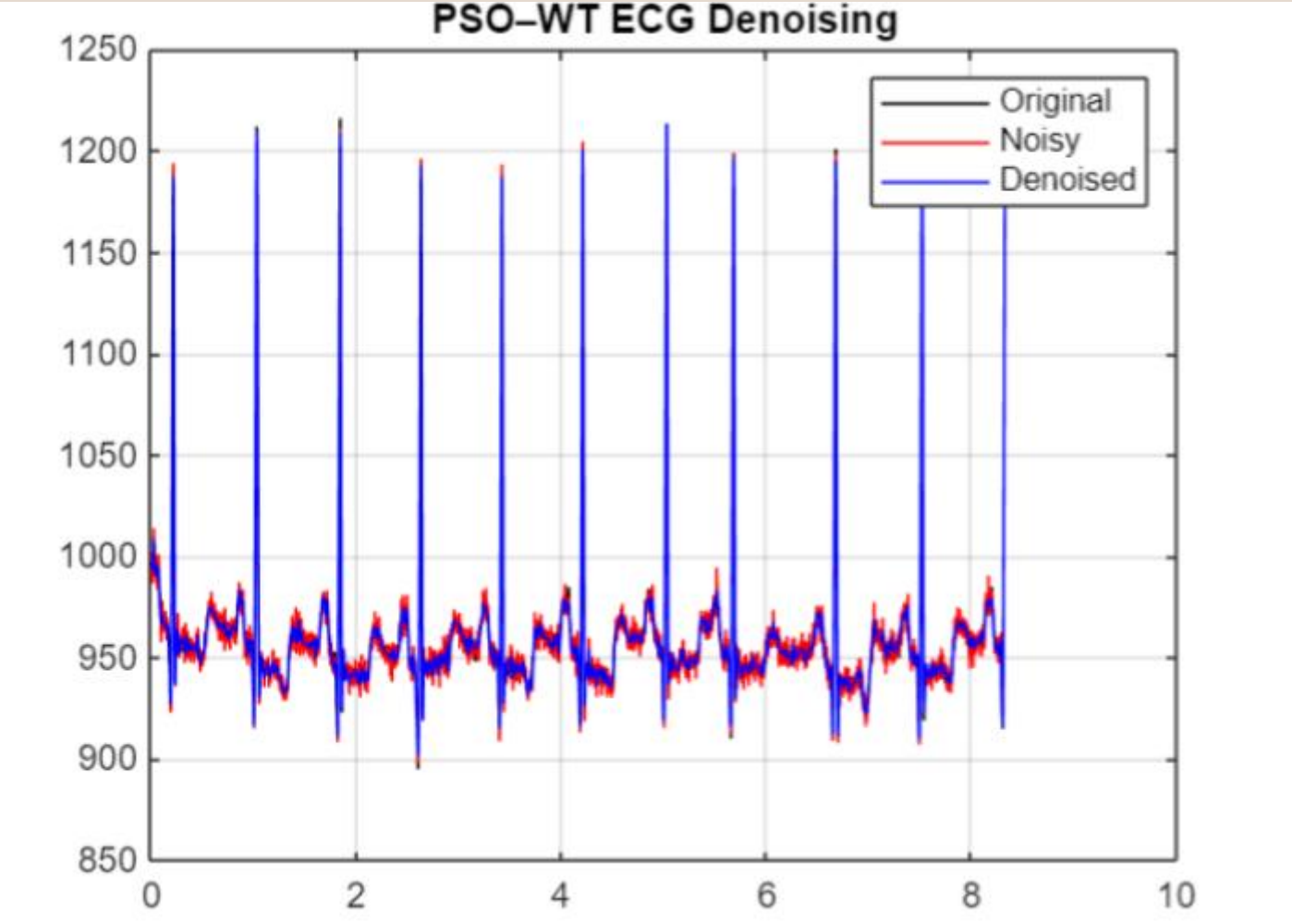
Position Update

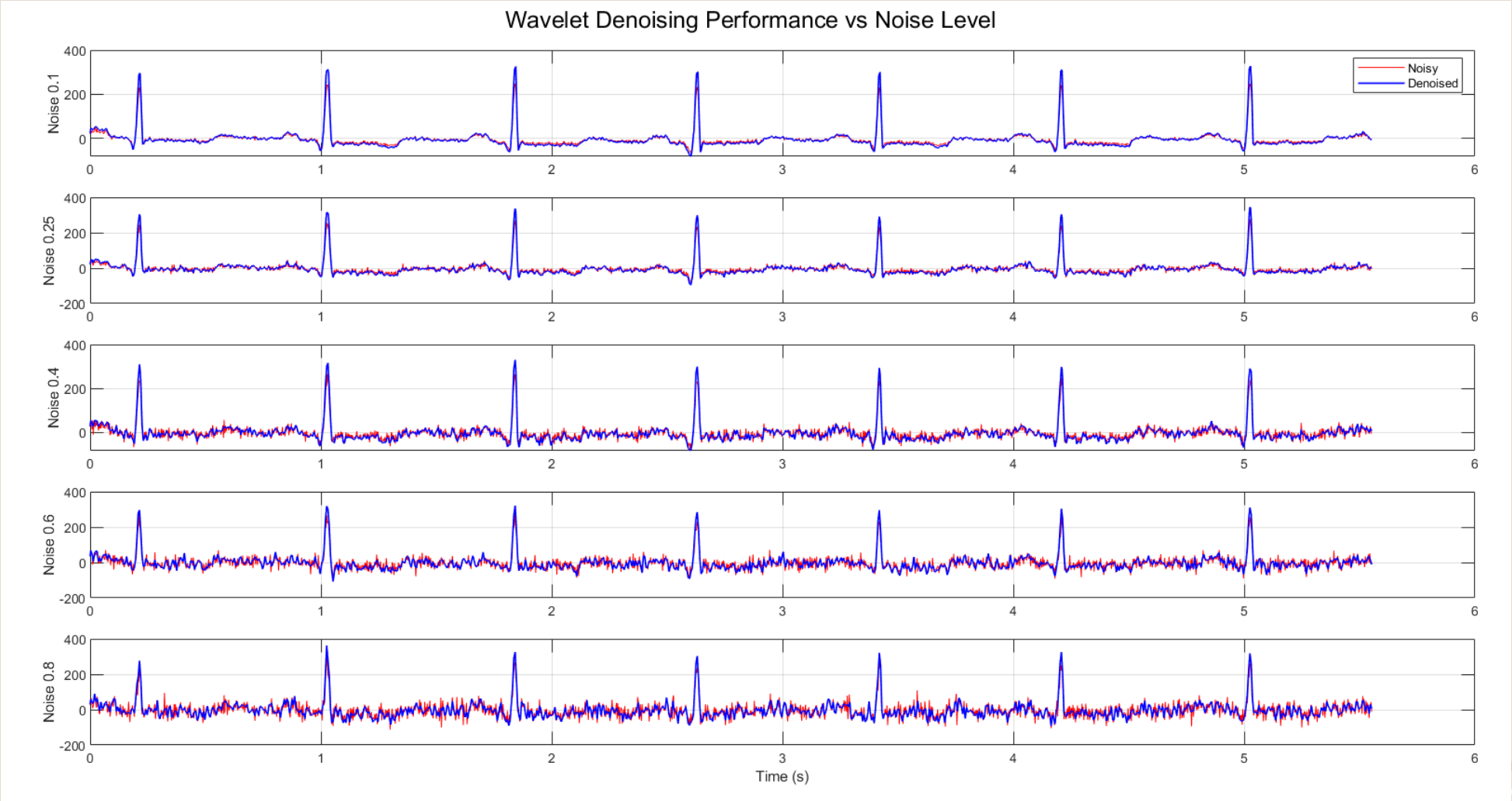
Particle.L=round(L+velocity)

L=4+2.4=6.4
L=6

- Random Reset of Other Parameters
- We are changing only the Level(L) parameter,moving towards the gbest

RESULTS


PSO–WT ECG Denoising
Original
Noisy
Denoised
1250
1200
1150
1100
1050
1000
950
900
850
0
2
4
6
8
10


Wavelet Denoising Performance vs Noise Level
Noisy
Denoised
Noise 0.1
Noise 0.25
Noise 0.4
Noise 0.6
Noise 0.8
400
200
0
-200
0
1
2
3
4
5
6
Time (s)

Thank You